SINGAPORE DIMENSIONS • GRADE 2A • CHAPTER 4

Multiplication & Doubles: 2, 5 & 10 Facts

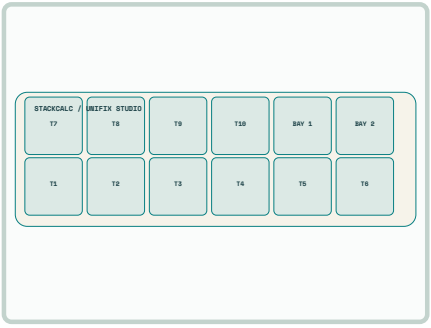
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.2.OA.C.4

Student Name: _____ Date: _____ **Model: SC-6** **Check Score: ___ / 4**

STUDENT SELF-CHECK RULE (NO CHEATING):
Build with blocks and solve on paper first! Use your SC-6 calculator ONLY to verify answers on the 4-line LCD screen.

STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

1. 2D Bench Blueprint



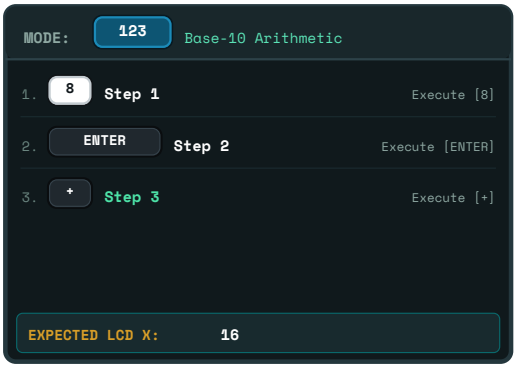
Kit: Unifix Tray

2. Desk Setup (Concrete -> Pictorial)

Physical Bench Setup:
Place double dominoes (4|4, 6|6) and build 2-column array tiles.

Sketch Model on Paper:
Draw an array grid: 2 rows of 6 dots.

3. Step-by-Step LCD Stack Trace (SC-6)



1 **Double 7: 7 ENTER +**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

2 **Double 9: 9 ENTER +**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

3 **3 groups of 5: [5] [ENTER] [3] [×]**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

4 **4 groups of 10: [10] [ENTER] [4] [×]**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

Station Diagnostic & Solution Key: Chapter 4

Multiplication & Doubles: 2, 5 & 10 Facts • Manipulative Setup, Diagnostic Rubric & Verification Flow

MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

Station Kit: Math Domino Studio & Counting Array Tiles (COUNT-01)

CAD Blueprint: unifix_tray

Target Device: SC-6 Elementary

OpenSCAD Source File:

Education/scad/counting_array_studio_parts.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

Recommended Filaments:

Oceanic Teal Case & 4-Color Array Tiles (PLA)

#	Math Problem Prompt	Expected Math Answer	Self-Check Keystrokes & RPN Stack
1	**Double 7: 7 ENTER +** Standard: CCSS.MATH.2.OA.C.4	14	[7] [ENTER] [+]
2	**Double 9: 9 ENTER +** Standard: CCSS.MATH.2.OA.C.4	18	[9] [ENTER] [+]
3	**3 groups of 5: [5] [ENTER] [3] [×]** Standard: CCSS.MATH.2.OA.C.4	15	[5] [ENTER] [3] [×]
4	**4 groups of 10: [10] [ENTER] [4] [×]** Standard: CCSS.MATH.2.OA.C.4	40	[1][0] [ENTER] [4] [×]

TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

Key Misconception to Target During Station Observation:

Distinguish 2 rows of 5 from 5 rows of 2.

How the Manipulatives & Calculator Foster Self-Correction:

Students solve with pencil and physical blocks first. When they self-check on the SC-6, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

Station Facilitation & Pacing Tip:

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

CLASSROOM SCORING & STATION MASTERY RUBRIC

Exemplary (18–20 pts):	Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack.
Proficient (14–17 pts):	Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting.
Developing (10–13 pts):	Understands problem concept but made computational slips; successfully self-corrected using calculator LCD.
Intervention (<10 pts):	Relied on calculator before solving on paper, or struggled with underlying physical manipulative models.